

Recap
Complex n th roots
Diagonalization
A brief history of complex numbers

Linear Algebra I

1. Complex roots 2. Diagonalization

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1 Recap

- Eigenvalues and eigenvectors of a matrix
- Complex numbers

2 Complex n th roots

- A motivating example
- Finding the fifth roots of 32

3 Diagonalization

- A 2x2 example
- The diagonalization theorem
- Examples/Non-examples

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Eigenvalues and eigenvectors of A

$$A\mathbf{x} = \lambda\mathbf{x} \text{ (and } \mathbf{x} \neq \mathbf{0}\text{)}$$

$$\iff$$

$$(A - \lambda I)\mathbf{x} = \mathbf{0} \text{ (and } \mathbf{x} \neq \mathbf{0}\text{)}$$

Which also implies

- λ is a solution of $\det(A - \lambda I) = 0$
- $\mathbf{x}$ is in $\text{Nul}(A - \lambda I)$

I will often abbreviate the words eigenvalues and eigenvectors to “evalues” and “evectors”

What we've seen so far

Complex numbers

- Some matrices have no real eigenvalues
- $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$
 - Addition, subtraction, multiplication, division
 - Complex conjugates $\overline{a + bi} = a - bi$
 - Absolute value aka modulus (=length) $|z| = \sqrt{z\bar{z}} = \sqrt{a^2 + b^2}$
 - Argument (=angle from real axis) $\arg(z) = \arctan \frac{b}{a}$
- Every complex polynomial has a solution in $\mathbb{C}$
 - Even the real ones
 - Every matrix (real or complex) has an eigenvalue
 - E.g. $\lambda^2 + 1 = (\lambda + i)(\lambda - i)$ has solutions $\pm i$
 - Fact: Non-real roots of real polynomials appear as conjugate pairs

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Learning outcomes

In today's class, we will:

- Investigate n^{th} roots of a complex number, motivated by a question about eigenvalues (Section 5.5)
(Using a 5x5 example matrix)
- Describe the process of diagonalizing a matrix (Section 5.3)
(Using a 2x2 example with real eigenvalues)

Key takeaways today

The #1 takeaways from today's class

- Complex numbers have n distinct n^{th} roots!
(two square roots, three cube roots, four 4th roots, etc.)
(we will see how to find them all)

- If $P = [\text{e vectors of } A]$ and $D = \text{diag}(\text{e values of } A)$, then

$$AP = PD$$

(when P is invertible: $AP = PD \iff A = PDP^{-1}$)
(we say P diagonalizes A) (we call A diagonalizable)

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Example question

Find all of the (complex) eigenvalues of the matrix A given below.

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 32 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

We will compute the characteristic polynomial $\det(A - \lambda I)$ and solve the equation $\det(A - \lambda I) = 0$

Step 1: Compute the characteristic polynomial:

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 32 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \text{ and so } A - \lambda I =$$

$$\begin{bmatrix} -\lambda & 0 & 0 & 0 & 32 \\ 1 & -\lambda & 0 & 0 & 0 \\ 0 & 1 & -\lambda & 0 & 0 \\ 0 & 0 & 1 & -\lambda & 0 \\ 0 & 0 & 0 & 1 & -\lambda \end{bmatrix}$$

Expand along row 1.

$$\det(A - \lambda I) = (-\lambda) \begin{vmatrix} -\lambda & 0 & 0 & 0 \\ 1 & -\lambda & 0 & 0 \\ 0 & 1 & -\lambda & 0 \\ 0 & 0 & 1 & -\lambda \end{vmatrix} + 32 \begin{vmatrix} 1 & -\lambda & 0 & 0 \\ 0 & 1 & -\lambda & 0 \\ 0 & 0 & 1 & -\lambda \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

$$= (-\lambda)(-\lambda)^4 + 32(1)^4 = -\lambda^5 + 32$$

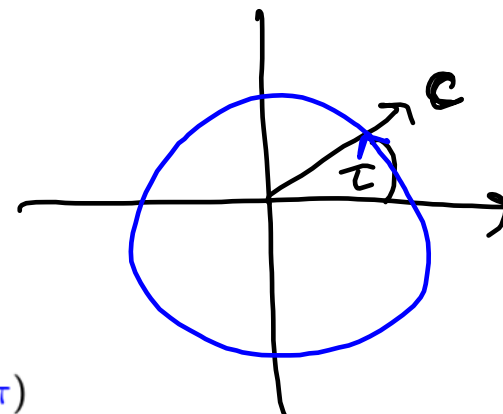
Step 2: Solve the characteristic equation

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 32 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \text{ and the characteristic equation of } A \text{ is } \lambda^5 = 32.$$

Question: How many solutions are there to the equation $\lambda^5 = 32$?

Let c be a complex number. To solve

$$z^5 = c$$



- ① Write z and c in polar (exponential) form

$$z = re^{i\theta}, \quad c = se^{i(\tau+2k\pi)}$$

(s and τ are given, we need to find the values r and θ)

τ

- ② Now compute z^5

$$z^5 = (re^{i\theta})^5 = r^5(e^{i\theta})^5 = r^5e^{i(5\theta)}$$

- ③ Equate lengths and angles:

$$r^5 = s \quad \text{and so} \quad r = \sqrt[5]{s} \quad (\text{the real positive 5th root of } s)$$

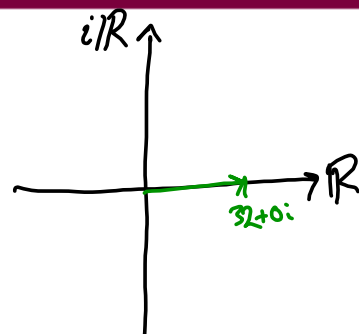
$$5\theta = \tau + 2k\pi \quad \text{and so} \quad \theta = \frac{\tau}{5} + \frac{2k\pi}{5}$$

The same procedure works for $z^n = w$

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Back to the example



$$\lambda^5 = 32$$

① $\lambda = re^{i\theta}, \quad 32 = se^{i(\tau+2k\pi)} = 32e^{i(0+2k\pi)}$

② $\lambda^5 = r^5 e^{i(5\theta)}$

③ Now equate lengths and arguments

$$r^5 = 32 \quad r = \sqrt[5]{32} = 2$$

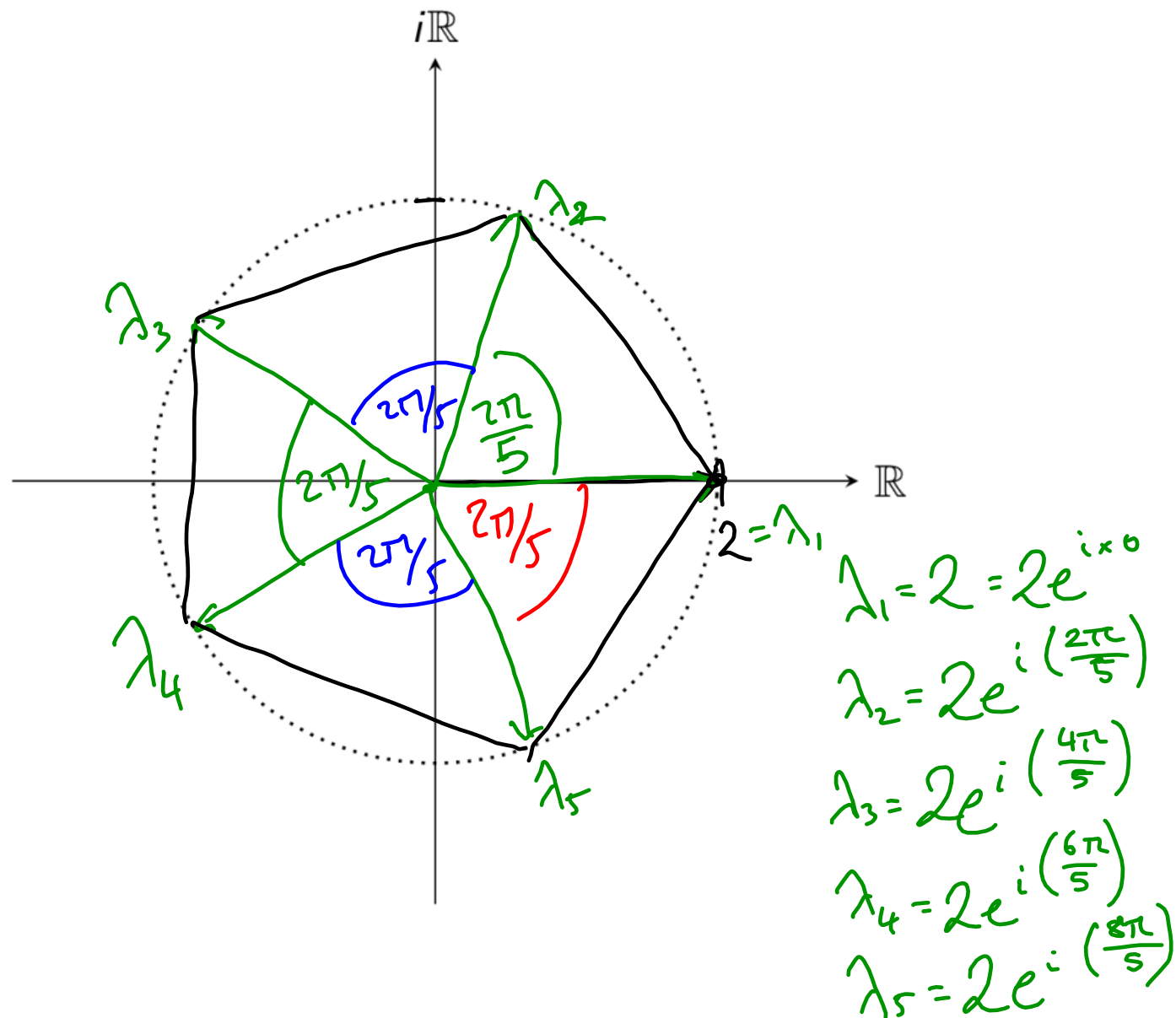
$$5\theta = 0 + 2k\pi, \quad \theta = \frac{0}{5} + \frac{2k\pi}{5} = \frac{2k\pi}{5}$$

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LAI Week 11

k	-1	0	1	2	3	4	5
θ	$-\frac{2\pi}{5}$	0	$\frac{2\pi}{5}$	$\frac{4\pi}{5}$	$\frac{6\pi}{5}$	$\frac{8\pi}{5}$	2π

Let's draw the solutions:



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In a nutshell

If

$$z^n = se^{i\tau}$$

then

- The length of z is the real positive n th root of s
- The argument of z is $\frac{\tau}{n} \pm \frac{2k\pi}{n}$

(This gives five distinct complex roots— they all lie equally spaced around a circle of radius $\sqrt[n]{s}$)

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Find the e vectors and evalues of C

$$C = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Find the e vectors and evals of C

$$C = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \quad C - \lambda I = \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix}$$

Solution:

$$|C - \lambda I| = \begin{pmatrix} 2-\lambda \\ \lambda-2 \end{pmatrix}^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3)$$

Eigenvalues are 1 and 3

$$\text{Nul}(C - 1I) = \text{span} \left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\} \leftrightarrow \left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\} \text{ is a basis for the eigenspace with eval 1}$$

$$\text{Nul}(C - 3I) = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$$

A matrix which diagonalizes C

Let P be a matrix whose columns are independent vectors of C . $P = \begin{bmatrix} -1 & 1 \\ 1 & 1 \\ \vec{v}_1 & \vec{v}_2 \end{bmatrix}$

Let D be a diagonal matrix whose diagonal entries are the corresponding values.

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}.$$

$$CP = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} -2+1 & 2+1 \\ -1+2 & 1+2 \end{bmatrix}$$

$$CP = \begin{bmatrix} -1 & 3 \\ 1 & 3 \end{bmatrix} \quad \swarrow \text{The same!}$$

$$PD = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ 1 & 3 \end{bmatrix}$$

$$\begin{aligned}
 CP &= C[\bar{v}_1 \ \bar{v}_2] \\
 &= [C\bar{v}_1 \ C\bar{v}_2] \\
 &= [\lambda_1 \bar{v}_1 \ \lambda_2 \bar{v}_2]
 \end{aligned}$$

common picture of matrix multiplication

eigenvectors of C

(diagonal on the right $\rightarrow$ scale the columns)

$$PD = [\bar{v}_1 \ \bar{v}_2] \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = [\lambda_1 \bar{v}_1 \ \lambda_2 \bar{v}_2]$$

If $P = [\bar{v}_1 \ \dots \ \bar{v}_n]$ is a matrix of vectors of A ,

then $AP = PD$ for $D = \text{diagonal eigenvalues}$

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Diagonalization in general

Theorem 5 from Section 5.3

The $n \times n$ matrix A may be written as $A = PDP^{-1}$ for a diagonal matrix D if and only if A has n independent eigenvectors.

In fact, when $A = PDP^{-1}$ with D a diagonal matrix, then P must be a matrix of eigenvectors of A , and D must be the corresponding eigenvalues.

Definition: *diagonalizable*

In the situation above, we say that A is *diagonalizable* and that P *diagonalizes* A

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Exercises to test your understanding

- Consider the matrices $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$.

Show that one of these matrices is diagonalizable and the other is not.

(Hint: Using the diagonalization theorem, a 2x2 matrix is diagonalizable if and only if it has two linearly independent eigenvectors)

- What does the diagonalization theorem tell us about the eigenvectors of a diagonal matrix?
(Hint: which matrix diagonalizes every diagonal matrix?)

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A (Very) Short History of Complex Numbers

The following timeline of imaginary numbers is for your interest only (non-examinable).

- **Cardano (1500s):** Encountered $\sqrt{-1}$ while solving cubics; used it reluctantly
- **Bombelli (1500s):** First coherent rules for $a + bi$ arithmetic
- **Descartes (1600s):** Coined the term “imaginary numbers” (dismissively)
- **Euler & de Moivre (1700s):** Made complex numbers central in analysis and trigonometry; $e^{i\theta} = \cos \theta + i \sin \theta$
- **Gauss (1800s):** Gave the modern viewpoint — complex plane, geometric meaning, and rigorous use

Complex numbers weren't invented all at once — they gradually became indispensable